Number Theory — problem 6.36

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$\gcd(n, n+1) = 1$, $\gcd(2n-1, 2n+1) = 1$, $\gcd(2n, 2n+2) = 2$, $\gcd(a, b) = \gcd(a, a+b)$, $\gcd(5a+3b, 13a+8b) = \gcd(a, b)$.